

Counting

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THE SUM RULE(union of Disjoint set)

If a task can be done either in one of n_1 ways or in one of n_2 ways, where none of the set of n_1 ways is the same as any of the set of n_2 ways, then there are $n_1 + n_2$ ways to do the task.

More formally: If sets $A_1, A_2, \dots, A_n$ are pairwise disjoint (no element appears in more than one set), then:

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n|$$

Key Requirement: Mutual Exclusivity

- The crucial condition is that the options must be **mutually exclusive** (disjoint) — you cannot count the same outcome twice.

Example 1

A restaurant offers 3 chicken dishes or 4 vegetarian dishes. How many ways can you choose a item?

Example 2

Suppose that either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?

Example 3

A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

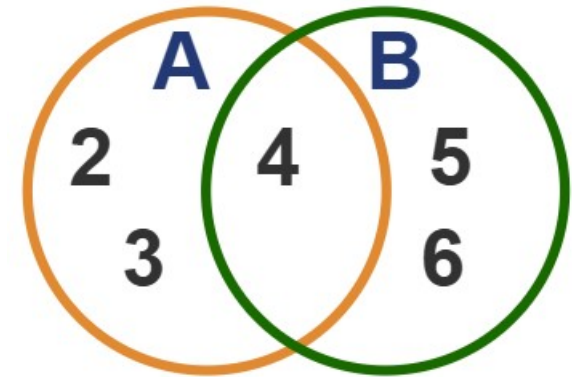
Example 4

If we role 2 dice one green and another is purple , how many ways there are to get a sum of 7 or 11?

THE SUBTRACTION RULE

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

$$|A \cup B| = |A| + |B| - |A \cap B|$$



In a class of undergraduate CSE students, 43 have taken Discrete mathematics , 52 have taken Linear Algebra, while 20 have taken both. No student has taken any other courses. How many students are there?

How many bit strings of length eight either **start with a 1 bit** or **end with the two bits 00**?

A computer company receives **350 applications** from college graduates for a job planning a line of new web servers. Suppose that **220** of these applicants majored in **computer science**, **147** majored in **business**, and **51 majored both** in computer science and in business. How many of these applicants majored neither in computer science nor in business?

THE DIVISION RULE

There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w , exactly d of the n ways correspond to way w .

Remark : There are n/d ways to do a task if:

- It can be done in n total ways,
- And for every actual way of doing the task, there are d repeated versions (duplicates) among those n ways.

Suppose that an automated system has been developed that counts the legs of cows in a pasture. Suppose that this system has determined that in a farmer's pasture there are exactly **572** legs. How many cows are there in this pasture, assuming that each cow has **four** legs and that there are no other animals present?

Suppose **100** cookies are distributed equally to a group of kids. Each kid receives **4** cookies. How many kids are there?

How many different ways are there to seat **four people** around a circular table, where two seating's are considered the same when each person has the same left neighbor and the same right neighbor?

The Pigeonhole Principle

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

Example

Examples 1–3 show how the pigeonhole principle is used.

EXAMPLE 1 Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays. 

EXAMPLE 2 In any group of 27 English words, there must be at least two that begin with the same letter, because there are 26 letters in the English alphabet. 

EXAMPLE 3 How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

THE GENERALIZED PIGEONHOLE PRINCIPLE

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

If **50** buildings are colored with **7** different colors then at least how many buildings will have the same color ?

Find the minimum number of teachers in a college to be sure that 4 of them are born in the same month.

30 Peoples in a party , at least how many people have birthday in the same month?

Minimum How many bicycles must be colored with **7** colors so as to guarantee minimum of **4** cycles have same color?

How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

What is the minimum number of students required in a discrete mathematics class to be sure that **at least six** will receive the same grade, if there are five possible grades, A, B, C, D, and F?

Permutations(Arranging Objects in Order)

Basically Permutation is an arrangement of objects in a particular way or order. While dealing with permutation one should concern about the selection as well as arrangement. In Short, ordering is very much essential in permutations. In other words, the permutation is considered as an ordered combination.

- An ordered arrangement of r elements of a set is called an **r-permutation**.

For example, if we have three letters, say A, B, and C, the possible permutations.

ABC	ACB	BAC
BCA	CAB	CBA

Representation of Permutation

$$P(n, k)$$

$$P_k^n$$

$${}_n P_k$$

$${}^n P_k$$

$$P_{n,k}$$

Permutation Formula

Permutation formula is used to find the number of ways to pick 'r' things out of 'n' different things in a specific order, and replacement is not allowed.

If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r -permutations of a set with n distinct elements.

If n and r are integers with $0 \leq r \leq n$, then $P(n, r) = \frac{n!}{(n - r)!}$.

How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Suppose that there are **eight runners** in a race. The winner receives a gold medal, the second-place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties?

Suppose that a saleswoman has to visit **eight different cities**. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

How many permutations of the letters **ABCDEFGH** contain the string **ABC** ?

How many **3-letter** words can be formed using the letters from the word "**FABLE**"?

In how many ways can a set of **five letters** be selected from the English alphabet?

How many 4 letter “words” can you make from the letters a through f, with no repeated letters?

Combinations

The combination is defined as “An arrangement of objects where the order in which the objects are selected does not matter.” The combination means “Selection of things”, where the order of things has no importance.

For example, if we want to buy a milkshake and we are allowed to combine any 3 flavours from Apple, Banana, Cherry, and Durian, then the combination of Apple, Banana, and Cherry is the same as the combination Banana, Apple, Cherry.

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An r-combination of elements of a set is an unordered selection of r elements from the set. Thus, an r -combination is simply a subset of the set with r elements.

Representation

- The number of r-combinations of a set with n distinct elements is denoted by $C(n, r)$.

- $\binom{n}{r}$

Formula

The number of r -combinations of a set with n elements, where n is a nonnegative integer and r is an integer with $0 \leq r \leq n$, equals

$$C(n, r) = \frac{n!}{r! (n - r)!}.$$

$$C(n, r) = \frac{n!}{r! (n - r)!} = \frac{n(n - 1) \cdots (n - r + 1)}{r!}.$$

Relation between Permutation and Combination

Theorem : ${}^n P_r = {}^n C_r \cdot r!$

Corresponding to each combination of ${}^n C_r$, we have $r!$ permutations because r objects in every combination can be rearranged in $r!$ ways.

Proof:

$$\begin{aligned} {}^n P_r &= {}^n C_r \cdot r! \\ &= [n! / r!(n-r)!] \cdot r! \\ &= n! / (n-r)! \end{aligned}$$

Theorem: ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

Proof:

$$\begin{aligned} {}^nC_r + {}^nC_{r-1} &= \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} \\ &= \frac{n!}{r(r-1)!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)(n-r)!} \\ &= \frac{n!}{(r-1)!(n-r)!} \left[\frac{1}{r} + \frac{1}{(n-r+1)} \right] \\ &= \frac{n!}{(r-1)!(n-r)!} \left[\frac{n+1}{r(n-r+1)} \right] \\ &= \frac{(n+1)!}{(r)!(n+1-r)!} \\ &= {}^{n+1}C_r \end{aligned}$$

Let n and r be nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n - r)$.

Proof: From Theorem 2 it follows that

$$C(n, r) = \frac{n!}{r! (n - r)!}$$

and

$$C(n, n - r) = \frac{n!}{(n - r)! [n - (n - r)]!} = \frac{n!}{(n - r)! r!}.$$

Hence, $C(n, r) = C(n, n - r)$.

How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?

How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?

A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission (assuming that all crew members have the same job)?

Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?

Find the number of subsets of the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ having 3 elements.